Syllabus CS6620 : Fundamentals of Cloud Computing (Online) - Summer A 2026

Description

Covers fundamentals of cloud computing, including virtualization and containers, distributed file systems and object stores, infrastructure as a service platforms, open source cloud platforms, key big data platforms, and topics in data center scale systems. Combines classroom material delivered via lectures, readings from literature, student presentations, and a semester-long software project

Course Objectives

- Understand the utility, concepts and purpose behind by cloud computing services
- Gain an appreciation of available unix utilities and containers for administration and their role in cloud computing
- Analyze various different component solutions that could go into architecting cloud solutions
- Build and deploy an applications in multiple different ways

General information

Instructor/Professor	**Aanchan Mohan**
Instructor email	aa.mohan@northeastern.edu
Recorded lectures posted by (two topics at a time)	Tuesday 2PM Pacific Time
Weekly Mock Interviews with TAs (schedule will be made available)	Tuesday Wednesday Thursday
Weekly exercises due by	Monday 5pm Pacific Time
Professor Office Hours	Friday 2pm-5pm Pacific Time booked via this link or by appointment (please email me)
Teaching Assistants (TAs)	Jingsi Zhang (zhang.jings@northeastern.edu) Aarushi Kaushik (kaushik.aar@northeastern.edu) Wendi Zhong (zhong.wend@northeastern.edu) Shurui Liu (liu.shur@northeastern.edu) Hongjing Ren (ren.hong@northeastern.edu) Shiqi Yang (yang.shiqi2@northeastern.edu)

NOTE: This is a compressed format of a course being delivered asynchronously. All students are expected to view any/all videos and attend mock interviews as scheduled. You should allocate up **to 20 hours per week** for this course.

Resource(s)

- https://learntocloud.guide/
- AWS Certified Solutions Architect Study Guide with 900 Practice Test Questions, 4th Edition by David Clinton and Ben Piper : available with your NU login on O'Reilly
- AWS Academy Learner Lab : https://awsacademy.instructure.com/courses/166236
- Online resource links to various content on Canvas

Course Particulars

- Discussion Board: https://piazza.com/northeastern/summer2026/cs6620
  Some class exercises will ask you to post your thoughts on a topic on the discussion board. Additionally, this online discussion board may be used to clarify your question about assignments and discuss general questions. When possible, students should ask related questions through this portal instead of sending individual emails to the instructor. Please do NOT post your programming code or solutions there. If you have a specific question about your solution, please ask help from the instructor, TA, or approved tutors in their offices.
- Teams and Email Policy: I will be unable to respond to you personally on MS Teams, just because of the high volume of messages I receive. If you do not receive a response, then consider that I have not read it. The best way to communicate with me is through a message on Piazza or via e-mail.

Topic schedule (subject to adjustment)

Material release	**Topics**	**Project**
Wed May 6, 2026	T1: Introduction to cloud computing, type I and type II hypervisors	Project for CS6620
Tue May 12, 2026	T2: Linux commands and remote access. Introduction to ssh. grep, rsync, scp T3: Introduction to networking - The different networking protocols, subnets, CIDR blocks, the different kinds of networks, DNS and firewalls.	
Tue May 19, 2026	T4: Automation with github actions - CI/CD pipelines and AWS Lambda T5: Containers, dependency management. Container registries.	Project team member names due
Tue May 26, 2026	T6: Microservices vs Monoliths T7: Linux permissions and groups, introduction to identity and access management (IAM); AWS S3	Project proposal slides due
Tue June 2, 2026	T8: Cloud Security AWS Virtual Private Cloud and security groups T9: Private subnets, NAT Gateways, stateful vs stateless connections Infrastructure as code - Introduction to Terraform	
Tue June 9, 2026	T10: Observability and monitoring T11: Production deployments	Project demo video 1 due on Monday June 8
June 15, 2026	Final Q and A week : Cloud architecture projects with system design aspects	Project demo video 2 due on Monday June 15

Course assessment

- Weekly exercises and lab modules: 35%
    - Pre-topic Exercise 1 : Grades awarded as points - 10%. The lowest assignment of the Exercise 1 group will be dropped.
    - Per-topic Exercise 2: Grades awarded as points - 10%. The lowest assignment of the Exercise 2 group will be dropped.
    - Per-topic Exercise 3: Grades awarded as points - 15%. The lowest assignment of the Exercise 2 group will be dropped.
- Mock Interviews : 30%
    - One mock interview/week for the following weeks (5 total mock interviews) : Weeks of May 11, May 18, May 25, June 1, June 8.
    - Mock interviews will be with TAs. In case you are unable to make it to a mock interview please look at the **Mock Interview Conduct** section for the rescheduling policy.
    - Each mock interview is worth 6% of your final grade (6% x 5 = 30%).
- Team final project - 35%
    - Team formation - 2.5%
    - Team Proposal - 7.5%
    - Project video report 1 - 10% (Individual grade)
    - Project video report 2 - 10% (Team grade)
    - Final Q and A - 5% (Team grade) - The style will be in the mock-interview format. This Q and A will also be with TAs, with the professor attending some of them.

Final grades

A	93.00 – 100.00 %
A-	90.00 – 92.99%
B+	86.00 – 89.99%
B	82.00 – 85.99%
B-	77.00 – 81.99%
C+	73.00 – 76.99%
C	69.00 – 72.99%
C-	65.00 – 68.99%
F	Less than 65%

Grading:

- **Weekly submissions:** Each topic (T1-T11) comes in most cases with 3 exercises - Exercise 1, Exercise 2 and Exercise 3.
    - Exercise 1: Consists of engaging with pre-topic content and answering questions. The lowest assignment of the Exercise 1 group will be dropped.
    - Exercise 2: Consists of engaging with topic content created by the instructor and completing a short lab or pen and paper exercise. The lowest assignment of the Exercise 2 group will be dropped.
    - Exercise 3: Consists of larger lab exercises on AWS Learner Labs that put some of the topic contents into practice. The lowest assignment of the Exercise 3 group will be dropped.

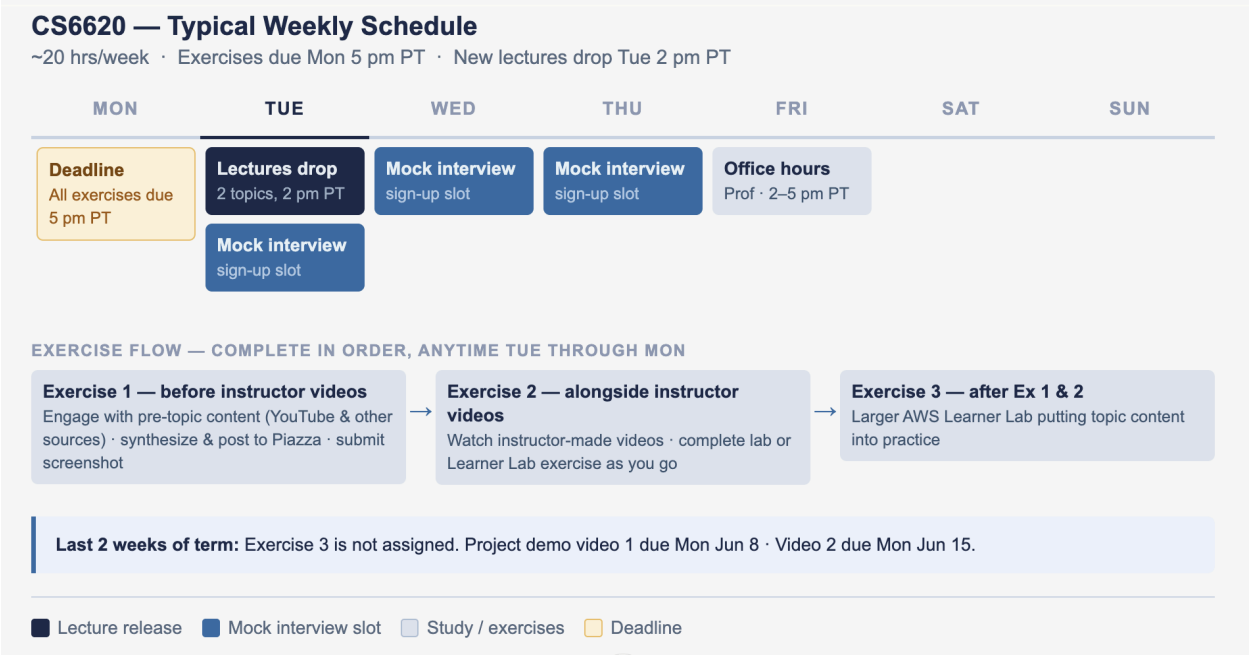
  Each exercise is graded on the following rubric:

    - ➔ Complete : 5 points - All questions answered completely and adequately
    - ➔ Partially complete: 2.5 points - Some questions answered adequately, but not all questions answered with complete justification and context.
    - ➔ Inadequate: 0 points - Questions are answered incomplete with little or no justification

- **Mock interviews:** Students will sign up, two at a time, for a 30 minute mock interview slot with a TA weekly. The mock interview will be conducted via MS Teams. The TA will interview two students at a time with questions alternating between students. Each mock interview slot is 30 minutes. The goal here is for the students to spend their time learning from each other (and from the TA). The questions will be related to the content of the exercises and the concepts covered in the topic and labs for the previous week.

  Each mock interview is graded on the following rubric:
    - ➔ Complete : 5 points - All questions answered completely and adequately
    - ➔ Partially complete: 3.5 points - Some questions answered adequately, but not all questions answered with complete justification and context
    - ➔ Signing up and showing up: 2 points!
    - ➔ No-show: 0 points

Typical weekly schedule (as an image)



Course policies:

Accommodations: The goal is that every student should be able to participate in this course. If you require any special accommodations, let me know immediately so that we can work out appropriate arrangements. Students who have disabilities who wish to receive academic services and/or accommodations should visit the Disability Access Services (DAS) (https://disabilityaccessservices.sites.northeastern.edu/) or call (844) 688-6287. If you have already done so, please provide your letter from the DAS to the instructor early in the semester to arrange those accommodations.

Student Expectations:

1. Respect should be shown in all communications and interactions with faculty, staff, industry, peers, and all others on campus. This includes respecting the preferred methods and response times of faculty and staff.
2. Your time in mock interviews will be devoted to small groups, and Team Based Learning activities. In Computer Science, we seldom get anything right on the first try. We see how an attempt turned out, and we try again. Often, any solution to a problem has a series of trade-offs, all of which must be considered! Our activities will reflect this approach as well; so please be prepared to present and discuss your solutions, even if you're not sure that your answer is "right"!
3. When you come to an interview, we ask that you be fully present. No phones are permitted in the interviews. Your camera must be on at all times. Please be respectful of your fellow students and teaching team by participating attentively and non-disruptively.
4. To create and preserve an atmosphere that optimizes teaching and learning, all participants share a responsibility in creating a civil and non-disruptive forum for the discussion of ideas. Students are expected to conduct themselves at all times in a manner that does not disrupt the group. Your comments to others should be constructive and free from harassing statements.

5. In order for this course to be a meaningful learning experience, you will need to come to each interview well-prepared, with all assigned readings and videos complete, as well as your individual work finished to the best of your ability. This emphasis on pre-interview work is the reason why our interviews are only 30 minutes each week. If you do not complete the pre-interview work, you will have a hard time following and contributing, which will make it that much harder for you to successfully complete the course.

Late work:

- Exercises: Late work will not be accepted for credit. For any project deliverables any submission after the deadline will be assessed at 50%. More than a day late will receive no credit.
- Mock Interviews: See the **Mock Interview Conduct** section below for the rescheduling and no-show policy.

Mock Interview Conduct

NOTE: Working cameras and microphones are required for this course. Mock interviews are conducted via MS Teams and your camera must be on at all times during mock interviews. **Mock interviews will be recorded.**

To create and preserve an atmosphere that optimizes teaching and learning, all participants share a responsibility in creating a civil and non-disruptive forum for the discussion of ideas. Students are expected to conduct themselves at all times in a manner that does not disrupt teaching or learning.

Specifically, all students will do the following when they enter a mock interview:

- Turn off ALL notifications on your phone and your computer
- Put your phone away (e.g. in your bag) so that it is not visible to you
- On your computer, close all apps that you do not need to be sharing

Your comments to others must be constructive and free from harassing statements. You are encouraged to disagree with other students and the instructor, but such disagreements need to respectful and be based upon facts and documentation, rather than prejudices and personalities. The TA reserves the right to interrupt conversations that deviate from these expectations. Repeated unprofessional or disrespectful conduct may result in a lower grade or more severe consequences.

Rescheduling and No-Show Policy: Mock interviews will be with TAs. In case you are unable to make it to a mock interview hosted by a TA, you may request one during the instructor's office hours via the booking link or email. **This has to be scheduled the week of the particular mock interview and not before or after. Otherwise the grade will be a "No show grade."**

AI Policy: Students will be working closely with AI tools. Each student has a Claude subscription from Northeastern University. We will often question what an AI tool gives as a suggestion for the exercises we do. It is important to emphasize that these tools should be used to support your efforts, *not replace* them. While AI can help, your unique intellectual contributions are crucial to the success of your project. We invite you to actively explore and experiment with AI tools, and we encourage you to share innovative ways to leverage these tools with your peers. In particular, we expect to discuss exactly when these tools fail, and the ramifications of using them recklessly!

Academic Integrity: Students are encouraged to use the Internet to enhance their learning. But outright copying without citing sources does not allow you to synthesize your learning and use this valuable time in your life effectively. You are also absolutely allowed to talk about high level details with your peers. Learning is never effective in isolation. All work is individual work unless an assignment states that a

question consists of group work. If the professor has concerns about your work your professor will set up a 1-1 meeting with you to discuss concerns about the assignment. The professor expects you to answer the questions to the best of your ability. **The professor reserves the right to ask you to verbally explain the reasoning behind any answer or code that you turn in and to modify your project grade based on your answers.**

Penalties for academic dishonesty include:

- Filing a report with the Office of Student Conduct and Conflict Resolution (OSCCR) (https://osccr.sites.northeastern.edu/academic-integrity-policy/) which would then bar you from registering co-ops or teaching and research assistantship positions on campus.
- Being awarded a 0 on your assignment.
- Repeated cheating will lead to failing the class which means you will receive an F grade.

Examples of academic dishonesty:

- Submitting a copy of the work done by another student, with or without their knowledge.
- Submitting work that was primarily found on the web or provided by someone else outside of this class.
- Submitting work by anybody who took this course in the past whether the course was here at Northeastern or at another campus or institution.
- Providing or receiving significant help to another student on an assignment.

If you are unsure or have a question about this policy just ASK!

Title IX Policy : Northeastern University and its faculty are committed to creating a safe and open learning environment for all students. If you or someone you know has experienced discrimination (including discrimination based on sex, gender, gender identity, gender expression, sexual orientation, pregnancy or pregnancy related condition, race, religion, national origin, disability status, veteran status etc.), or sexual violence (including sexual harassment, sexual assault, dating/domestic violence, or stalking), please know that help and support are available. Northeastern strongly encourages all members of the community to take action, seek support, and report incidents of discrimination, harassment, and sexual violence to the Office for University Equity and Compliance (OUEC) through the Online Reporting Form.

Please be aware that faculty members are Mandatory University Reporters who are required to disclose information about alleged discrimination, harassment, and sexual violence to the OUEC. If the OUEC receives a report, a member of their office will reach out to offer information about available rights, support resources and pathways towards a resolution as a member of the campus community. Community members are not required to respond to this outreach.

If you, or another community member you know wishes to speak to a confidential resource who does not have this reporting responsibility, please contact any of the following confidential resources. These confidential resources are not required to report allegations of discrimination to the University without your signed release.

- Find@Northeastern: Offers 24/7 mental health support via phone at 877.233.9477 (in the U.S.) or +1.781.457.7777 (outside the U.S.).
- Sexual Violence Resource Center: The SVRC provides confidential, trauma-informed support services to Northeastern students who have experienced any form of sexual violence (i.e., sexual assault, sexual harassment, sexual exploitation, domestic/dating violence, and/or stalking). Request services online at bit.ly/svrequestform.
- Confidential Resource Advisor: The CRA provides confidential, restorative informed support services to Northeastern students who have been accused of sexual or identity-based harm. Request services online at bit.ly/svrequestform.

- Pregnant / parenting students: Please know that the OUEC, housing the University's Title IX Coordinator, can work with students who are pregnant and/or parenting to ensure they have equal access to education programs and activities. For additional support, please contact OUEC (ouec@northeastern.edu).

Please visit ouec.northeastern.edu for a complete list of reporting options and support resources both on- and off-campus and contact the OUEC (ouec@northeastern.edu) at any time.

Feedback

Your opinions are very important to me. All students are strongly encouraged to use the Teacher Rating and Course Evaluation (TRACE) system, at https://www.northeastern.edu/trace/ , to complete your course evaluations. A reminder about TRACE should arrive via email about two weeks before the end of the course.

Wellness and Mental Health Support

As a graduate student, you may experience a range of challenges including significant stress, difficult life events, mood changes, excessive worry, or problems with eating and/or sleeping. If you or anyone you know is struggling, we strongly encourage you to seek support.

Northeastern University provides several services and resources to support the overall wellness of students. To access support, you can book a Wellness Consultation with a Wellness Program Specialist. During this session, you can discuss your concerns and receive guidance on the next steps, along with access to resources that promote mental health and overall well-being.

Students in need of immediate support can access Find@Northeastern for free 24/7 mental health support at 855.229.8797 (Canada) and +1.781.457.7777 (International) or Here2Talk, a free 24/7 counselling service for all post-secondary students in BC at: 604-642-5212 or toll free at 1-877-857-3397

Learning is most easily accomplished when you are physically and emotionally at your best. If you run into difficulties and need assistance, I encourage you to contact me during my office hours, reach out before or after class, or send me an email. I will do my best to support your success during the term. This includes identifying concerns I may have about your academic progress or wellbeing through a Wellness Consultation Referral. Through this process, I can connect you with the campus Wellness Program Specialist who offers support and assistance getting back on track to success. Only the Wellness Program Specialist can access any concerns I may identify, and a referral does not affect your academic record.